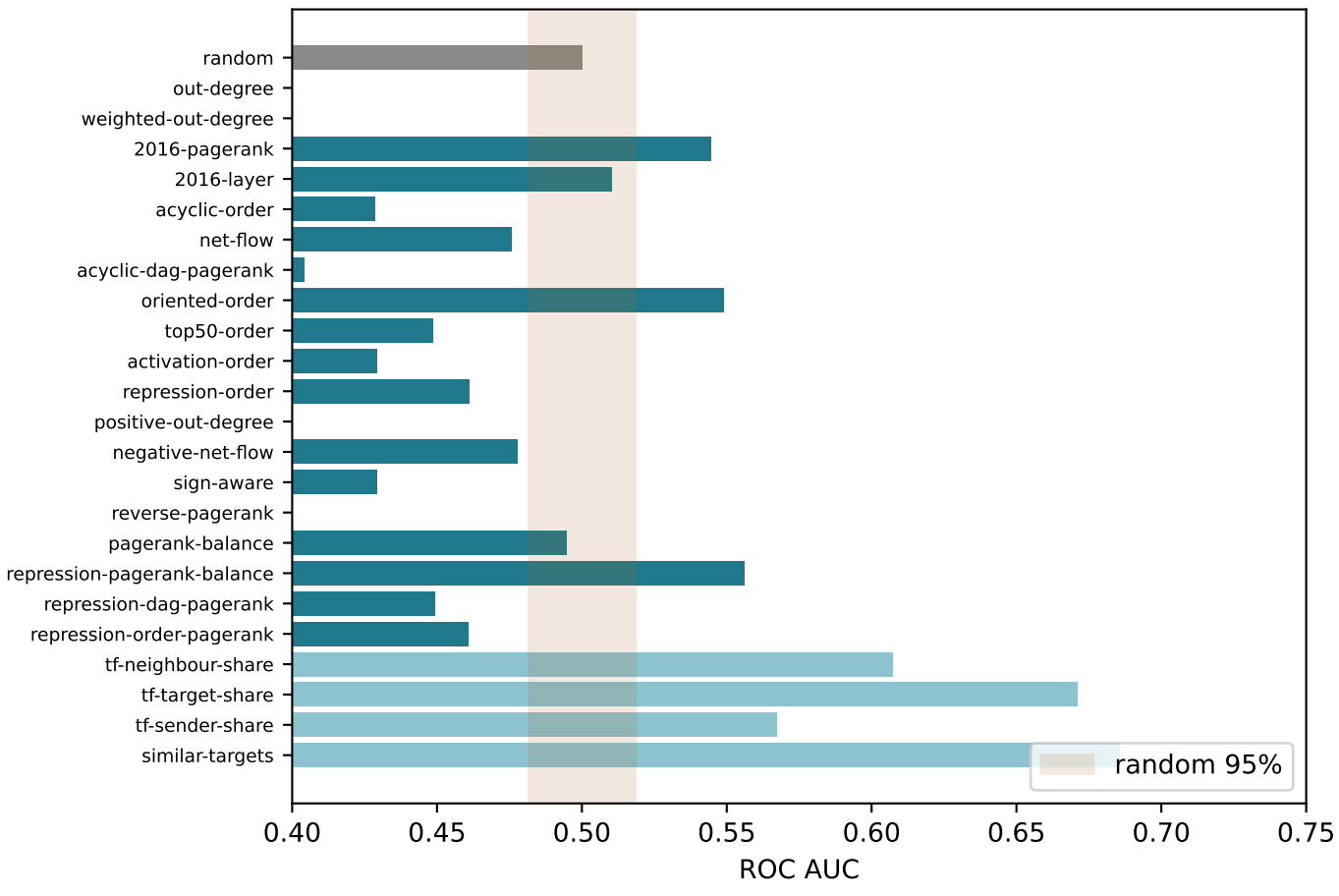
Regulator ranking against reference 'lambert-2018'

2026-11-10 · 11,173 genes ranked, 994 in lambert-2018-tfs.txt · top k = 100

Pipeline	Reference genes in top 100	AUC	p top-k	p AUC	AUC vs best baseline (95%)	p vs baseline
random	8.9 (4–15)	0.500 (0.482–0.518)				
out-degree	5	0.391	0.949	1.000	+0.000 (+0.000, +0.000)	
weighted-out-degree	5	0.389	0.949	1.000	-0.001 (-0.003, -0.000)	0.022
2016-pagerank	6	0.544	0.891	< 0.001	+0.154 (+0.125, +0.181)	< 0.001
2016-layer	7	0.510	0.799	0.136	+0.119 (+0.099, +0.139)	< 0.001
acyclic-order	5	0.429	0.949	1.000	+0.038 (+0.019, +0.058)	< 0.001
net-flow	5	0.476	0.949	0.993	+0.085 (+0.071, +0.100)	< 0.001
acyclic-dag-pagerank	4	0.404	0.979	1.000	+0.013 (+0.007, +0.020)	< 0.001
oriented-order	4	0.549	0.979	< 0.001	+0.158 (+0.132, +0.183)	< 0.001
top50-order	11	0.449	0.263	1.000	+0.058 (+0.034, +0.083)	< 0.001
activation-order	13	0.429	0.105	1.000	+0.038 (+0.017, +0.060)	< 0.001
repression-order	3	0.461	0.995	1.000	+0.070 (+0.054, +0.086)	< 0.001
positive-out-degree †	3	0.389	0.995	1.000	-0.002 (-0.006, +0.003)	0.514
negative-net-flow †	5	0.478	0.949	0.989	+0.087 (+0.072, +0.103)	< 0.001
sign-aware †	4	0.429	0.979	1.000	+0.038 (+0.030, +0.048)	< 0.001
reverse-pagerank †	4	0.396	0.979	1.000	+0.005 (+0.003, +0.008)	< 0.001
pagerank-balance †	3	0.495	0.995	0.736	+0.104 (+0.080, +0.126)	< 0.001
repression-pagerank-balance †	6	0.556	0.891	< 0.001	+0.165 (+0.143, +0.187)	< 0.001
repression-dag-pagerank †	5	0.449	0.949	1.000	+0.058 (+0.043, +0.073)	< 0.001
repression-order-pagerank †	5	0.461	0.949	1.000	+0.070 (+0.054, +0.084)	< 0.001
tf-neighbour-share ‡	32	0.607	< 0.001	< 0.001	+0.216 (+0.195, +0.237)	< 0.001
tf-target-share ‡	62	0.671	< 0.001	< 0.001	+0.280 (+0.254, +0.306)	< 0.001
tf-sender-share ‡	23	0.567	< 0.001	< 0.001	+0.176 (+0.155, +0.198)	< 0.001
similar-targets ‡	69	0.686	< 0.001	< 0.001	+0.295 (+0.267, +0.323)	< 0.001

Random: mean and 95% range over random orderings. p: share of random orderings scoring at least as well. Grey bars: random and baselines. AUC vs best baseline: paired bootstrap over genes (1,000 resamples). †: designed after seeing these results; needs confirming on other data. ‡: uses known TFs: scored from other genes' labels in this reference (lighter bars).



Edge checks against curated TF→target pairs

Recall ×: share of curated pairs present as edges, over chance.
Positive/negative ×: share of edges of that sign that are curated pairs, over chance. Sign: agreement with the curated sign (chance).

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Graph	Pairs	Recall (×)	Positive ×	Negative ×	Sign (chance)
raw graph	30,436	0.053 (2.75×)	5.51×	1.31×	0.68 (0.62)
resolve_reciprocals → drop_sinks → prune_triangles	30,382	0.016 (1.38×)	3.81×	1.33×	0.30 (0.29)
resolve_reciprocals → drop_sinks → prune_triangles → greedy_layering	14,989	0.028 (0.94×)	4.17×	0.93×	0.27 (0.27)
absolute	30,436	0.053 (2.75×)			
absolute → minimum_feedback_order	30,063	0.040 (2.37×)			
orient_reciprocals ratio=2.0 → absolute	30,310	0.033 (2.05×)			
orient_reciprocals ratio=2.0 → absolute → minimum_feedback_order	30,299	0.028 (1.87×)			
top_outgoing k=50 → absolute	30,436	0.016 (5.92×)			
top_outgoing k=50 → absolute → minimum_feedback_order	30,112	0.011 (5.65×)			
keep_sign sign=1 → absolute	30,258	0.037 (5.48×)			
keep_sign sign=1 → absolute → minimum_feedback_order	29,266	0.029 (6.21×)			
keep_sign sign=-1 → absolute	20,290	0.025 (1.00×)			
keep_sign sign=-1 → absolute → minimum_feedback_order	19,125	0.025 (0.93×)			
keep_sign sign=-1	20,290	0.025 (1.00×)		1.00×	0.27 (0.27)

Pipeline steps

out-degree: Most outgoing edges (raw graph)

score out_degree, descending

Graph: 3,038,599 edges (2,013,864 negative), largest strong component 11,120.

weighted-out-degree: Largest total outgoing weight (raw graph)

score weighted_out_degree, descending

Graph: 3,038,599 edges (2,013,864 negative), largest strong component 11,120.

2016-pagerank: 2016 cleaning, lowest PageRank

resolve_reciprocals() → drop_sinks() → prune_triangles() → score pagerank, ascending

Graph: 1,815,396 edges (1,785,288 negative), largest strong component 10,825.

2016-layer: 2016 cleaning, earliest greedy layer

resolve_reciprocals() → drop_sinks() → prune_triangles() → greedy_layering() → score level, ascending

Graph: 1,815,396 edges (1,785,288 negative), largest strong component 10,825; DAG keeps 1,631,542 edges.

acyclic-order: Earliest in acyclic order

absolute() → minimum_feedback_order() → score level, ascending

Graph: 3,038,599 edges (0 negative), largest strong component 11,120; DAG keeps 2,591,369 edges.

net-flow: Outgoing minus incoming weight

absolute() → score net_flow, descending

Graph: 3,038,599 edges (0 negative), largest strong component 11,120.

acyclic-dag-pagerank: Reverse PageRank on the acyclic DAG

absolute() → minimum_feedback_order() → score dag_reverse_pagerank, descending

Graph: 3,038,599 edges (0 negative), largest strong component 11,120; DAG keeps 2,591,369 edges.

oriented-order: Pairs with a 2x stronger side only, earliest in acyclic order

orient_reciprocals(ratio=2.0) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 2,470,902 edges (0 negative), largest strong component 10,959; DAG keeps 2,347,856 edges.

top50-order: Top 50 outgoing edges per gene, earliest in acyclic order

top_outgoing(k=50) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 383,085 edges (0 negative), largest strong component 10,052; DAG keeps 270,045 edges.

activation-order: Activating edges only, earliest in acyclic order

keep_sign(sign=1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 1,024,735 edges (0 negative), largest strong component 9,729; DAG keeps 673,737 edges.

repression-order: Repressing edges only, earliest in acyclic order

keep_sign(sign=-1) → absolute() → minimum_feedback_order() → score level, ascending

Graph: 2,013,864 edges (0 negative), largest strong component 5,115; DAG keeps 1,985,600 edges.

positive-out-degree: Most positive outgoing edges

score positive_out_degree, descending

Graph: 3,038,599 edges (2,013,864 negative), largest strong component 11,120.

negative-net-flow: Negative outgoing minus incoming weight

score negative_net_flow, descending

Graph: 3,038,599 edges (2,013,864 negative), largest strong component 11,120.

sign-aware: Rank sum: positive out-degree and negative net flow

score sign_aware, descending

Graph: 3,038,599 edges (2,013,864 negative), largest strong component 11,120.

reverse-pagerank: Highest reverse PageRank (raw graph)